



AI Assessment & Academic Intelligence Sy

VedaAI – Full Stack Engineering Assignment

Role: Full Stack Engineer

Duration: 21 March (11:59PM)

Submission: GitHub Repository + Deployed link

Overview

Build an **AI Assessment Creator** based on the provided Figma designs.

The system should allow a teacher to:

- Create an assignment
- Generate a question paper using AI
- View the generated output

Extra Points for creativity. (Functionality)

This is the **Figma file**, which you have to implement.

Figma : [Figma Designs for AI Assessment Creator](#)

Core Features

1. Assignment Creation (Frontend)

Using the Figma designs, build a form with:

- File upload (PDF / text) (*optional*)
- Due date
- Question types
- Number of questions + marks
- Additional instructions

Requirements:

- Proper validation (no empty / negative values)
 - Use Redux or Zustand for state management
 - Websocket management
-

2. AI Question Generation

Requirements:

- Convert input → structured prompt
- Generate:
 - Sections (A, B, etc.)
 - Questions
 - Difficulty (easy / medium / hard)
 - Marks

Do not directly render LLM response.

3. Backend System

Stack:

- Node.js + Express (TypeScript)

Must include:

- MongoDB → store assignments & results
- Redis → caching / job state

- BullMQ → background jobs (generation, PDF)
- WebSocket → real-time updates

Flow:

1. API request
 2. Job added to queue
 3. Worker processes generation
 4. Store result
 5. Notify frontend
-

4. Output Page (Enhanced)

Display the generated question paper in a **structured and well-designed format**, inspired by the provided UI.

Required Elements:

Student Info Section

- Name (input line)
 - Roll Number (input line)
 - Section (input line)
-

Question Sections

- Group questions into sections (e.g., Section A, B, etc.)
- Each section should include:
 - Title
 - Instruction (e.g., "Attempt all questions")
 - Questions list

Each question must display:

- Question text
 - Difficulty tag (Easy / Moderate / Hard)
 - Marks
-

UX Expectations

- Clean, readable layout (similar to real exam papers)
 - Proper spacing and hierarchy
 - Mobile responsive
-

Bonus Features (Optional but High Signal)

- Download as PDF (proper formatting, not raw HTML print)
- Action bar (Regenerate)
- Highlight difficulty visually (badges/tags)

Avoid

- Rendering raw AI response
 - Poor formatting or misaligned sections
 - Single block of text without hierarchy
-

Tech Stack

Frontend:

- Next.js + TypeScript
- Redux / Zustand
- WebSocket

Backend:

- Node.js + Express
- MongoDB
- Redis
- BullMQ

AI:

- Any LLM (GPT / Claude / OSS)
 - Prompt structuring + parsing required
-

Submission

Submission Link : [Submit here](#)

1. GitHub Repo

- Clean code
- Setup instructions

2. README

- Architecture overview
- Approach

Bonus (Optional)

- PDF export
- Better caching
- Improved UI polish